

The solutions for this sheet must be submitted on Moodle by 30 April, 15:59. Afterwards you are assigned solutions of another student, which you have to grade (and upload to Moodle) until 11:59 of the following TUESDAY.

Exercise P4.1 – Randomized Algorithms

We want to solve the following problem: Given a graph $G = (V, E)$ and a natural number k , find a subset of edges $E' \subseteq E$ such that $G' = (V, E')$ is k -colorable. Describe a randomized algorithm that computes such a set E' with $|E'| > \frac{k-2}{k}|E|$. The algorithm should have linear expected runtime. That is, if X denotes the random variable that counts the number of elementary operations executed by the algorithm, then $\mathbb{E}[X]$ should be $\Theta(|V| + |E|)$.

Hint: You could proceed as follows: (i) find a linear-time algorithm that has a positive constant probability of finding such a set E' ; (ii) repeat this algorithm until it is successful; (iii) bound the expected number of repetitions needed.